

SkillMatchAI Internal Resume Skill Matching System Group 1 Amazon

Project Assignment 2
Due Date:
March 9, 2026

Presented by Ali Benrami, Swara
Saoji, Yash Baruah:
Project Manager:Yash

*Weekly Meetings held on thursday at 12:00-12:45pm
Microsoft teams*

Shared drive is Google

Messaging is via Discord

Team Profile Page



Ali Benrami
Morocco
Wylie, TX
Computer
Science

Student &
Intern

I enjoy attending
hackathons. I have
been to 3 so far.



Swara Saoji
India
Richardson, TX
Cognitive Science
AI/ML

Student and
Computing Lab
Associate

I get very
competitive during
board games.



Yash Baruah
Australia
Southlake, TX
Computer
Science

Student &
Intern

I enjoy researching
and building PCs

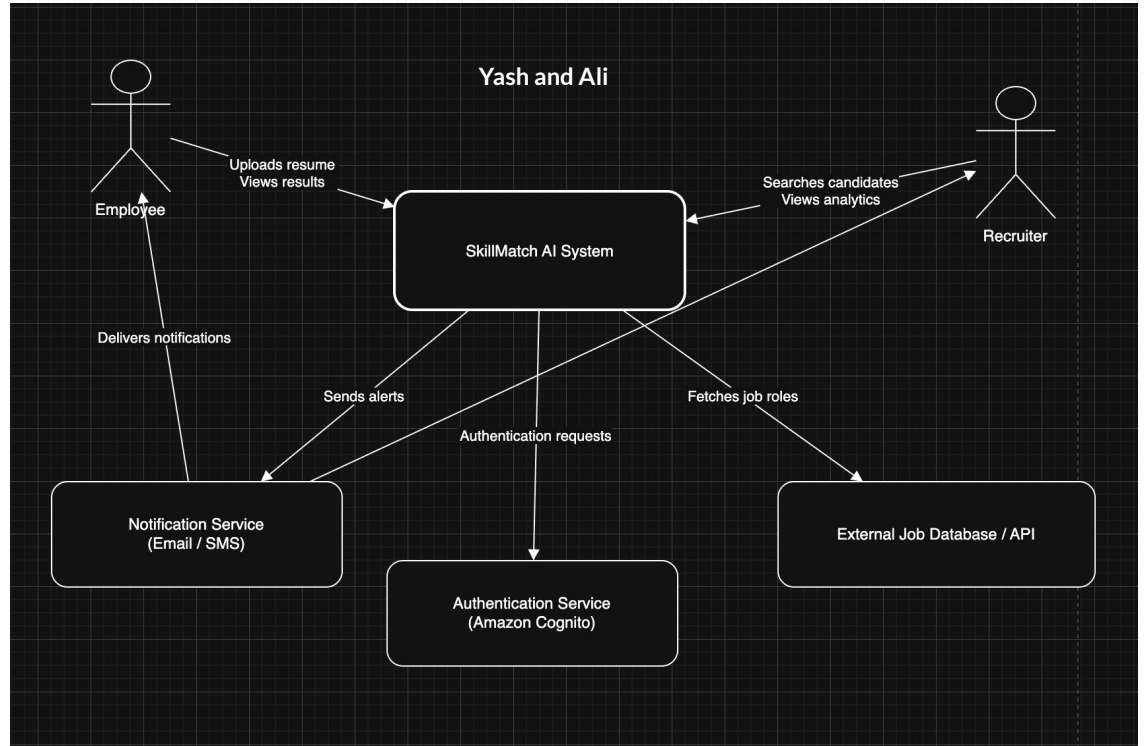
SOW Summary

- An AI system that analyzes employee resumes against internal job roles and recommends training paths.
- The company we expect to be providing this product to is Amazon
- This automation tool provides Amazon with a more robust and clean path for current employees to fill high priority roles as well as giving prospective employee other job opening that fit there experience
- We pick this system because we believe that now more than every it's hard to get hired but even harder to hire. Are project idea should help to alleviate those concern
- Brainstorming was done at are standard weekly meetup, and consisted of us bouncing ideas between each other.
- The key components of this project will be the resume extractor which converts resume content to a easily parsable file and the recommendation system which picks out the best possible candidates for any given role.

Architecture Overview Points

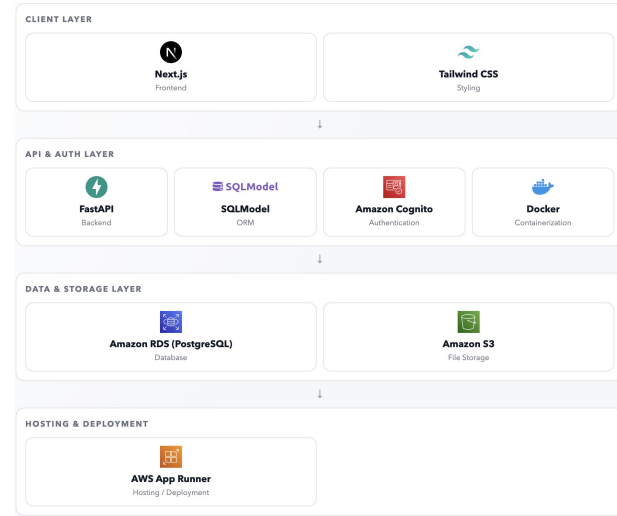
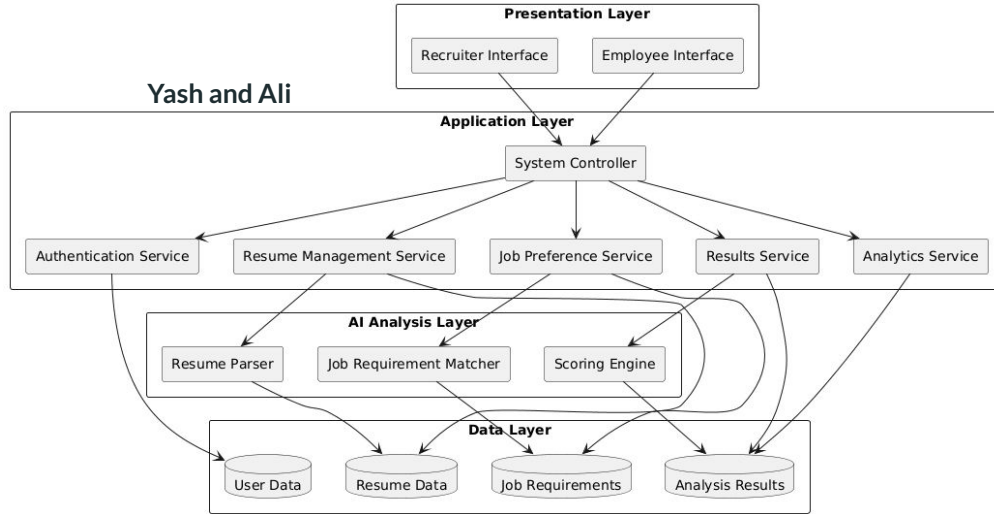
- The architecture of the system will be a layered system, it will consist of a frontend, backend, AI, and storage layers
- The system will be deployed on AWS cloud platform, which will ensure scalability and reliability of the system
- The system will be a web-based system, meaning users will interact with it using a frontend built using Next.js
- The backend will be built using FastAPI, which will be used to communicate with the AI services
- The system will consist of an AI module, which will be used to parse resumes and match skills
- The system will use Amazon RDS for storing structured data and Amazon S3 for storing files
- The system will use Amazon Cognito for user authentication

Context Diagram



System Architecture (Logical View) – Tech Stack

- System Architecture (Logical View) – Tech stack

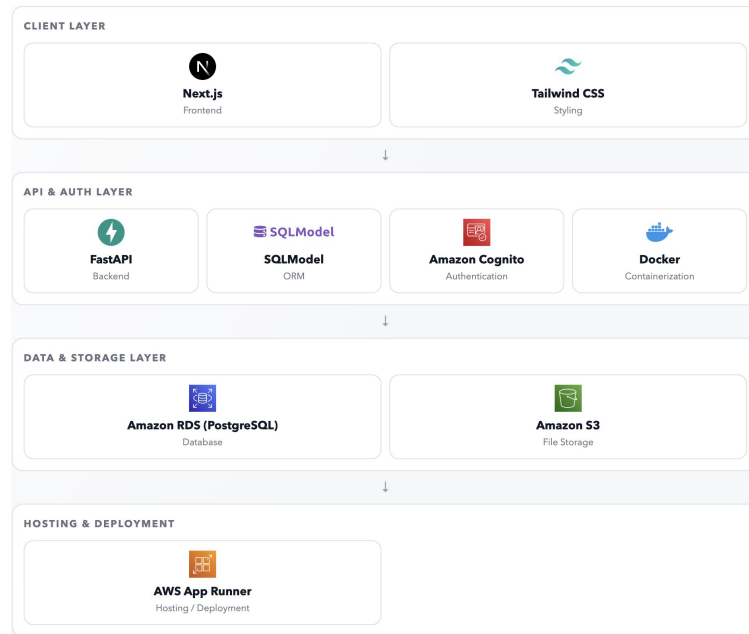


System Architecture (Physical View)

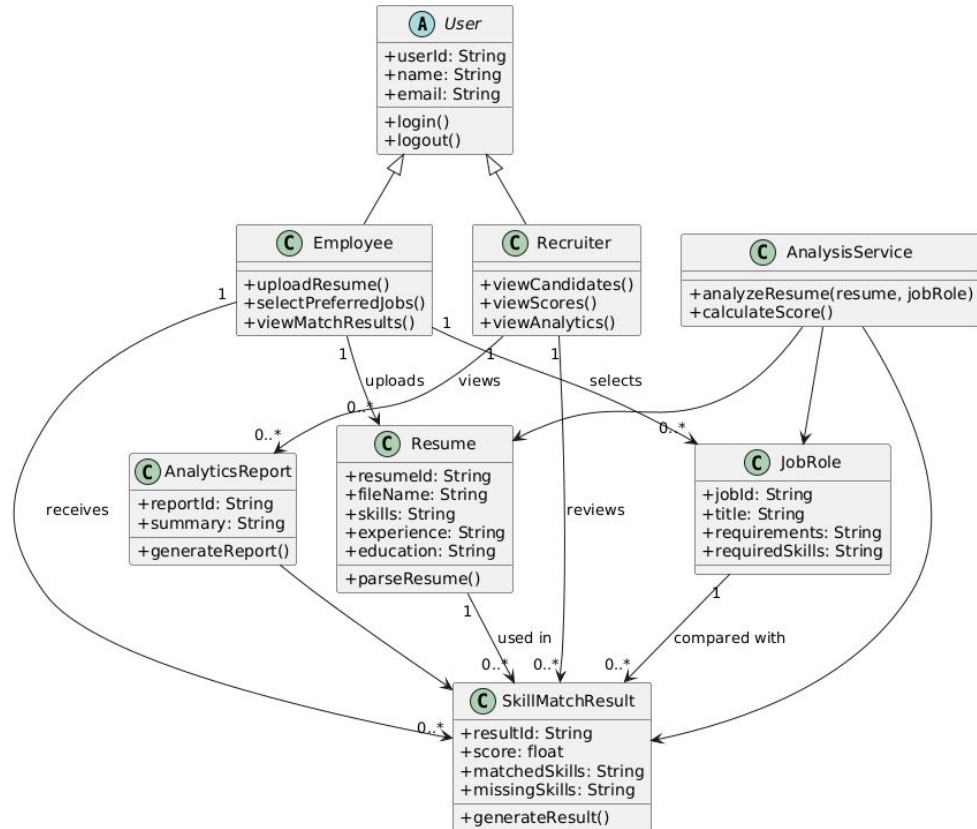


Tech Stack

- Use images of each technology you plan to use in coding, this can / will change for PA4
- Frontend: Next.js
- Styling: Tailwind CSS
- Backend: FastAPI
- Authentication: Amazon Cognito
- Database: Amazon RDS for PostgreSQL
- File Storage: Amazon S3
- ORM: SQLModel
- Containerization: Docker
- Hosting / Deployment: AWS App Runner



UML Class Diagrams



Checkpoint 3 Retrospective

Team Performance

- This team has done great work and has performed above expectation.
-

What Went Well

- We collaborate well together and get work done in a timely manner
- Everyone is easy to work with and is respectful to one another
- Ect

What Can Be Improved

- ?

Stakeholder Interview Questions & Requirements

Swara Saoji

1. Employees should be able to upload their resume in PDF or DOCX format.
 2. Employees should be able to drag and drop their resume into the upload area.
 3. After a resume is uploaded, the system should analyze the resume to identify the employee's skills and qualifications.
 4. Employees should be able to select a target internal role from a list of available roles.
 5. The system should compare the employee's skills with the requirements of the selected role.
 6. Employees should be able to view a skill match score showing how well their resume fits the selected role.
 7. Employees should be able to see which of their skills match the role requirements.
 8. Employees should be able to see which skills are missing for the selected role.
 9. The system should recommend learning resources or training options to help employees develop missing skills.
 10. Employees should be able to upload an updated resume.
1. What information should employees see after uploading their resume?
 2. How should the system display skill gaps to employees?
 3. What would make the results dashboard easy to understand?
 4. Would visual charts or text explanations be better for skill analysis?
 5. What additional features would make the system more useful?
 6. What information should be included in role skill requirements?
 7. How should skill gaps be communicated to employees?
 8. What insights would HR want from the system?
 9. How often should job role requirements be updated?
 10. How should the system recommend learning resources?

Stakeholder Interview Questions & Requirements

Yash Baruah

1. The system should require Amazon single sign-on before any internal user can access the platform.
 2. The system should assign access permissions based on user roles such as employee, recruiter, hiring manager, Learning and Development lead, and system administrator.
 3. The system should keep an audit log of sensitive actions such as logins, failed login attempts, recommendation generation, recruiter overrides, report exports, and administrative updates.
 4. The system should allow system administrators to view system health information, sync status, and queue backlogs from connected services.
 5. The system should notify administrators when a connected service, import, or data-sync job fails.
 6. The system should retry safe background jobs after temporary service failures and prevent partial or inconsistent updates.
 7. The system should show employees which skills match a target role, which skills are missing, and how important each missing skill is.
 8. The system should generate prioritized learning recommendations that are directly tied to the employee's missing skills and selected target role.
 9. The system should allow employees to save target roles, review previous analyses, and track progress after updating skills, training, or certifications.
 10. The system should provide Learning and Development teams with aggregated reporting on common skill gaps across departments, employee groups, and role families.
1. What authentication and access control model should SkillMatch AI use to protect internal hiring data?
 2. How should the system log and audit sensitive actions performed in the platform?
 3. What availability and recovery expectations should be defined for the platform?
 4. How should the platform handle failures in connected services or data-sync jobs?
 5. What browser and platform support should be expected for internal users?
 6. How should SkillMatch AI help employees understand why they are not a strong match for a target role?
 7. What kind of learning recommendations should the system generate to help employees become qualified?
 8. How should the system allow employees to track progress after receiving recommendations?
 9. What reporting would help Learning and Development teams improve workforce readiness?
 10. How should the platform prioritize learning recommendations when employees have many skill gaps?

User Stories / Non-Functional Requirements

Swara Saoji

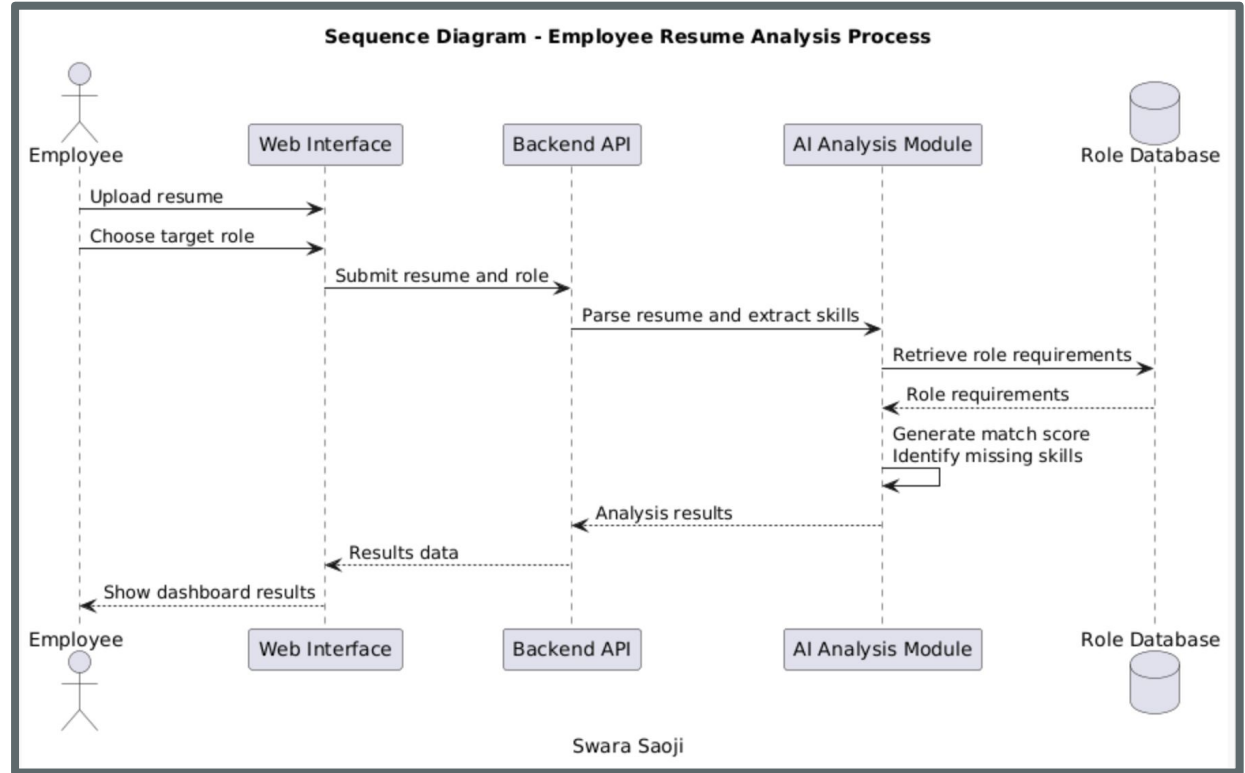
1. As an employee, I want to upload my resume so the system can analyze my skills.
 2. As an employee, I want to see how well my skills match a target role so I can understand my qualifications.
 3. As an employee, I want to track my analysis results over time to see how my skills improve.
 4. As an employee, I want to view a summary dashboard so I can quickly understand my skill profile.
1. The system should provide a simple interface that is easy for employees to use.
 2. The system should work on modern web browsers like Chrome, Safari, Edge
 3. The system should keep logs of system activity and errors.
 4. The system should allow system updates without major interruptions.

User Stories / Non-Functional Requirements

Yash

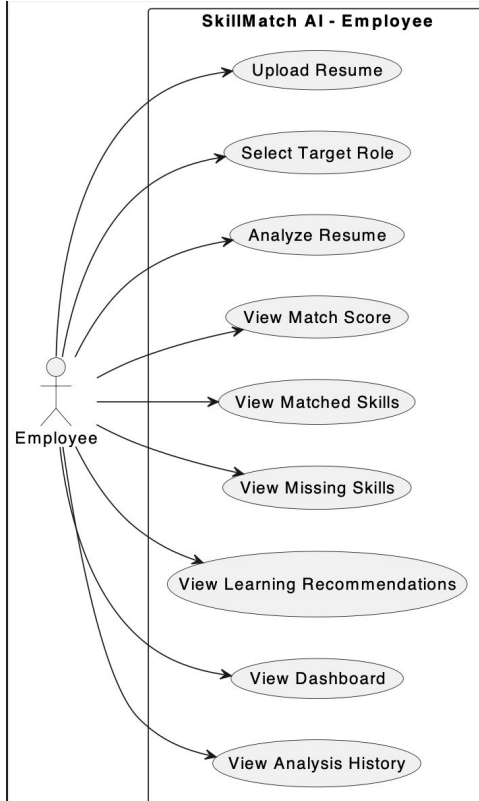
1. As a system administrator, I want users to sign in through Amazon SSO so that only approved internal users can access hiring data.
 2. As a system administrator, I want to monitor audit logs and integration status so that I can respond to failures before they affect users.
 3. As an employee, I want to see my missing skills and their importance for a target role so that I know what I need to improve.
 4. As a Learning and Development lead, I want to view common skill-gap reports and prioritized learning recommendations so that I can support workforce development better.
1. The system should maintain high availability during normal business hours and peak hiring periods.
 2. The system should protect audit logs from tampering and store each event with a timestamp and authenticated user identity.
 3. The system should work reliably on approved modern desktop browsers used by internal users.
 4. The system should isolate failures in connected services so that one integration problem does not bring down the full platform.

Sequence Diagram with overview Swara Saoji



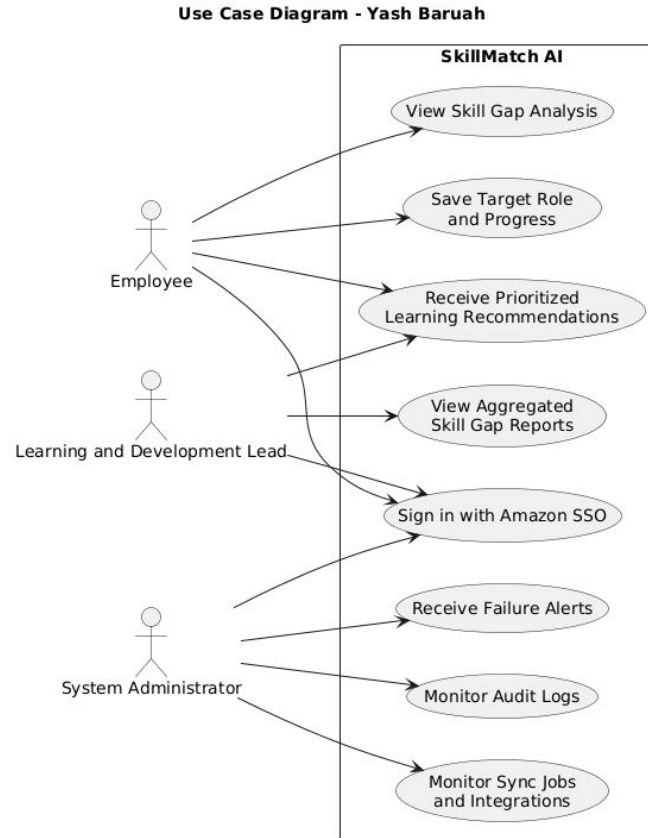
User Case Diagram with overview

Swara Saoji



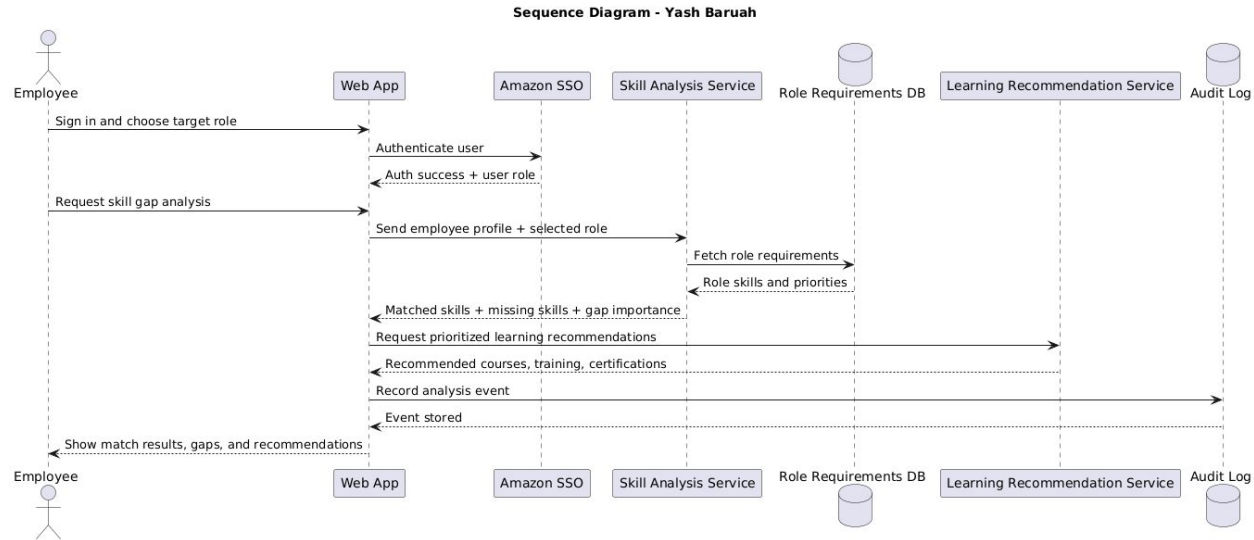
Sequence Diagram with overview

Yash Baruah



User Case Diagram with overview

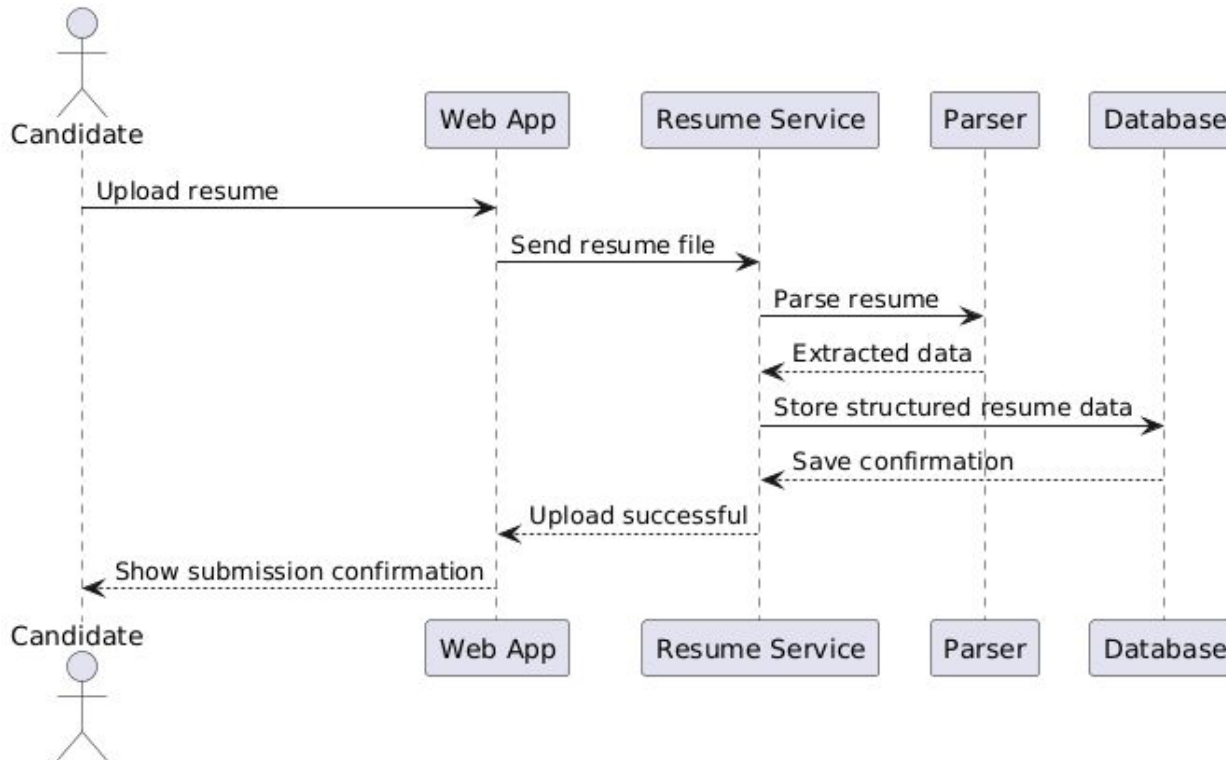
Yash Baruah



Sequence Diagram with overview

Ali Benrami

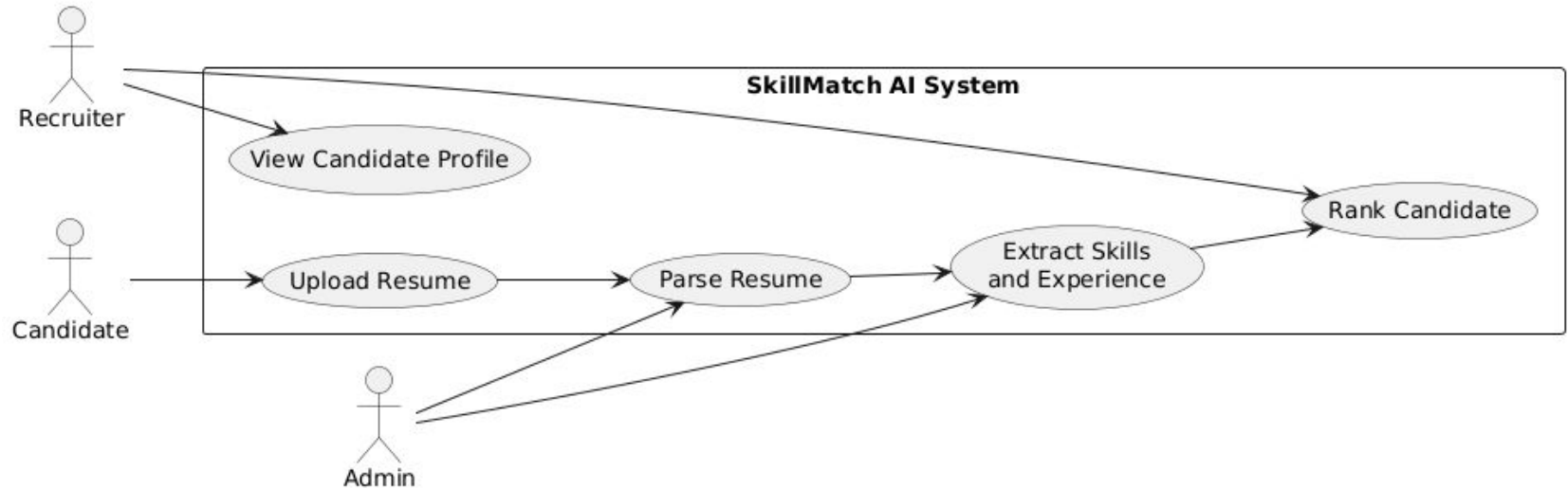
Sequence Diagram - Ali Benrami



Use Case Diagram with overview

Ali Benrami

Use Case Diagram - Ali Benrami



Wireframes

Swara Saoji

SkillMatch AI

Upload Your Resume



Drag & Drop Resume
OR

Browse Files

Target Role:

Analyze Resume

Supported formats: PDF, DOCX
Your data is securely processed

SkillMatch AI Dashboard

Notifications

Profile

Dashboard

Upload Resume

Skill Analysis

Learning Paths

Settings

Logout

Welcome, User!
Track your career skill growth and resume match progress.

Resume Analysis History

Data Scientist → 78% Match

ML Engineer → 72% Match

Software Engineer → 65% Match

Skill Progress

Python

Machine Learning

SQL

Deep Learning

Recommended Next Steps

- Learn Deep Learning
- Improve Cloud Computing
- Practice ML Projects

SkillMatch AI

Dashboard

Logout

Target Role: Data Scientist
Match Score: 78%

Matched Skills

- ✓ Python
- ✓ SQL
- ✓ Machine Learning
- ✓ Data Analysis

Missing Skills

- ✗ Deep Learning
- ✗ Docker
- ✗ Cloud Computing

Recommended Skills

- Deep Learning Specialization
- AWS Cloud Fundamentals
- Docker for Beginners

Improve Resume

Upload New Resume

Wireframes

Ali Benrami

SkillMatch AI

Screen 3: Learning and Progress Planner

Learning and Progress Planner

Requirement-aligned planning page that connects missing skills to prioritized learning resources and visible progress checkpoints.

Plan Summary

Target Role: Data Scientist

Priority Gaps: Cloud Platforms, Docker, Deep Learning

Goal Date: September 2026

Recommended Learning Resources

AWS Cloud Fundamentals
Mapped to Cloud Platforms

Docker for Data Teams
Mapped to Docker

Deep Learning Specialization
Mapped to Deep Learning

Progress Timeline

April

May

July

September

Start AWS module and complete baseline assessment.

Finish Docker lab and upload project evidence.

Update resume and rerun analysis.

Complete readiness review with manager.

Completed training and uploaded evidence update role readiness over time.

SkillMatch AI

Screen 1: Skill Passport

Skill Passport

Requirement-aligned employee profile showing verified skills, saved target roles, and analysis history for future career moves.

Navigation

Skill Passport

Role Matches

Analysis History

Learning Plans

Settings

Profile Summary

Avatar

Jordan Smith

Department: Business Intelligence

Preferred Role Family: Data and AI

Transition Ready

Saved Target Roles

Data Scientist
Fit 78% | Missing: Cloud Platforms

Analytics Engineer
Fit 81% | Missing: dbt Tooling

Verified Skills

Python

SQL

Analytics

Data Storytelling

Confidence from resume and training evidence

Analysis History

March 2026
Updated resume improved score by 6%

February 2026
Added certification evidence

SkillMatch AI

Screen 2: Role Comparison Board

Role Comparison Board

Employees compare internal roles by fit score, missing skills, and estimated readiness before selecting a target.

Data Scientist

Fit Score: 78%

MATCHED SKILLS

Python

SQL

Analytics Storytelling

MISSING SKILLS

Cloud Platforms

Docker

TIME TO READINESS

3 to 4 months

Choose Role

Analytics Engineer

Fit Score: 81%

MATCHED SKILLS

SQL

Data Pipelines

Reporting Systems

MISSING SKILLS

dbt Tooling

Warehouse Modeling

TIME TO READINESS

1 to 2 months

Choose Role

Decision Notes

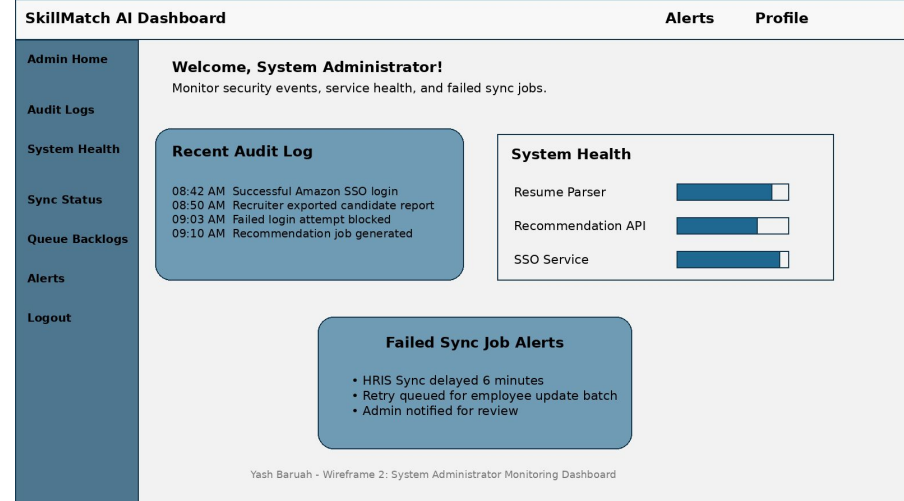
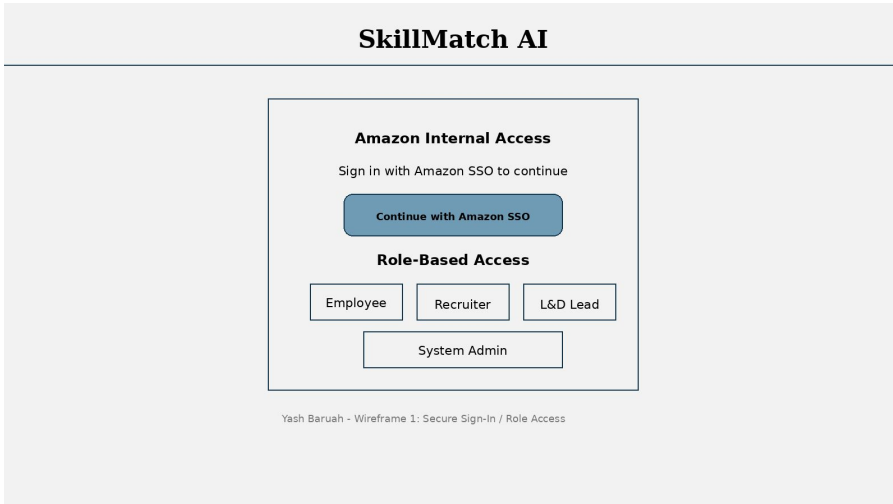
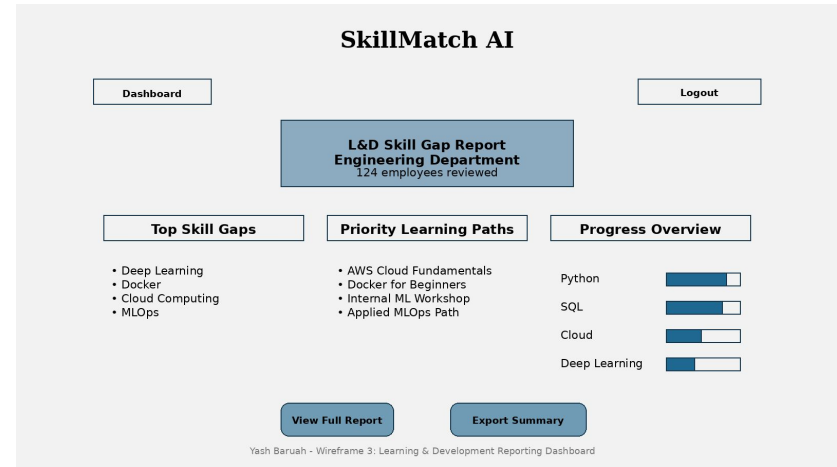
Closest next move based on current evidence:
Analytics Engineer

Recommended next step:
open the learning plan for the selected role.

Save Comparison

Wireframes

Yash Baruah



Architecture Overview Points

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Meeting Minutes – Project Assignment 1 and 2

Group Number: 01		Project Name: Project Group 1 aka	
Project Manager: Swara			
Meeting Date	Attendees	Absent	Topic Discussed
12-Feb	Swara S., Ali B., Yash B.		Reviewed Assignment 2 requirements, confirmed the SkillMatch AI project scope, and split the document into requirements, diagrams, wireframes, architecture, implementation, testing, and glossary sections. Established that the SED and presentation would both be updated from the shared Google Drive.
19-Feb	Swara S., Ali B., Yash B.		Completed stakeholder interview questions, user stories, and most functional and nonfunctional requirements. Discussed employee, recruiter, administrator, and Learning & Development workflows to keep the platform aligned with Amazon’s internal hiring use case.
26-Feb	Swara S., Ali B., Yash B.		Worked on system modeling deliverables including use case diagrams, sequence diagrams, site map planning, and wireframes. The team also reviewed the layered architecture direction so the user interface, backend services, AI analysis, and database responsibilities stayed clearly separated.
3-Mar	Swara S., Ali B., Yash B.		Finalized the remaining SED sections, including software testing, software evolution, and glossary updates, and reviewed the presentation for consistency with the SOW and requirements document. Confirmed that the testing plan would focus on functional, integration, and non-functional validation for reliability, security, and recovery behavior

Meeting Minutes – Project Assignment 3

Group Number: 01		Project Name: Project Group 1 aka	
Project Manager: Swara			
Meeting Date	Attendees	Absent	Topic Discussed
10-Mar	Swara S., Ali B., Yash B.		Worked on architecture diagrams and updated SED
17-Mar	Swara S., Ali B., Yash B.		Finalized context and physical diagrams, added external systems
24-Mar	Swara S., Ali B., Yash B.		Completed UML classes and presentation slides
31-Mar	Swara S., Ali B., Yash B.		Reviewed and finalized all deliverables

Who Delivered What?

Assignment	Deliverables	Who Delivered What?
1	SOW Team Profile Page Project Idea Slide Project Summary Slide SOW summary Meeting tracking	All All All Yash Ali Swara
2	Software Testing & Glossary of Terms Sow update & site map RTM Table & test case/requirements table SED	Yash Ali Swara All

Who Delivered What?

Assignment	Deliverables	Who Delivered What?
3	SED redesign Context Diagram Systems Arch (Logical) Systems Arch (Physical) Tech Stack UML Classes	Swara Yash + Ali Ali + Yash Swara + Yash Ali Ali + Swara
4		
5		